

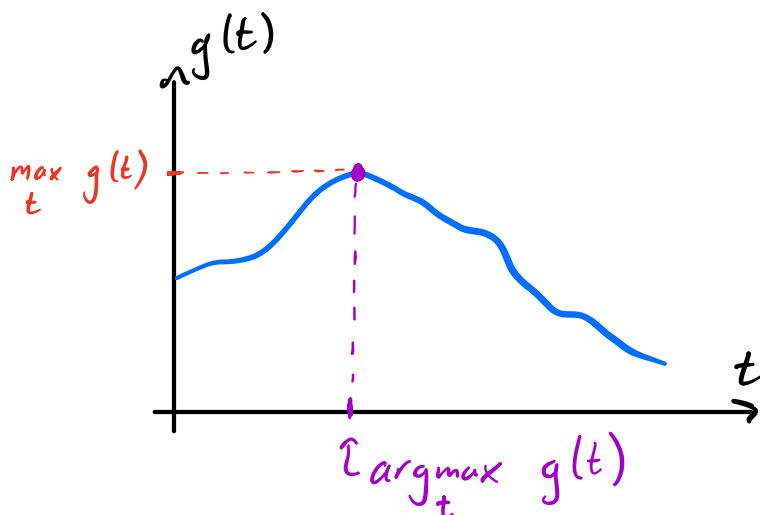
Maximum Likelihood Estimation:

So far, we have studied how to evaluate different point estimators, with no discussion of how to construct a point estimator.

Now, we get to the most common and useful technique of point estimation!

Preliminaries:

Def. (Argmax): Given a real-valued function $g(t)$, we say that $t_{\max} \in \arg\max_t g(t)$ if for all t , $g(t_{\max}) \geq g(t)$.

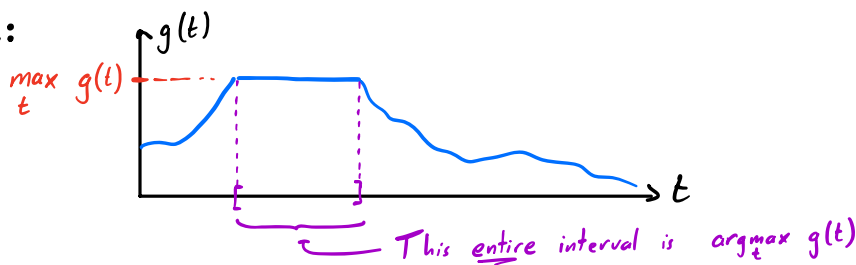


$\max_t g(t)$ = largest value of $g(t)$ we can get

$\arg\max_t g(t)$ = set of values of t for which $g(t)$ is as large as possible

Note: A function may have multiple argmaxes.

E.g.:



→ We will always work with cases where the argmax is unique

Note: $\arg\min_t g(t)$ and $t_{\min}$ are defined similarly.

Q: How do we find $\operatorname{argmax}_t g(t)$?

A: In this class, you will almost always be able to do the following:

- Compute the derivative of g with respect to t , getting $g'(t)$
- Solve for $t_{\max}$ by using $0 = g'(t_{\max})$

Note: Many details / assumptions are missing here... But this isn't a course in optimization. For our purposes, this will be enough, and is known as "solving the first-order conditions".

Note: $\operatorname{argmax}_t g(t) = \operatorname{argmax}_t \log(g(t))$, as long as $g(t) > 0$.

Moreover, $\operatorname{argmax}_t g(t) = \operatorname{argmax}_t a \cdot g(t) + b$, as long as $a > 0$.

However: $\operatorname{argmax}_t g(t) = \operatorname{argmin}_t (-1) \cdot g(t)$

Def. (Likelihood Function): Let $X_1, \dots, X_n$ be random variables with joint distribution parametrized by θ (or $\theta_1, \theta_2, \dots$), denoted $f(x_1, \dots, x_n; \theta)$ (or $p(x_1, \dots, x_n; \theta)$ if discrete). For some given realizations $x_1, \dots, x_n$, the likelihood function equals:

$$\mathcal{L}(\theta; x_1, \dots, x_n) = \begin{cases} f(x_1, \dots, x_n; \theta) & \text{if } X_1, \dots, X_n \text{ are cts.} \\ p(x_1, \dots, x_n; \theta) & \text{if } X_1, \dots, X_n \text{ are discrete} \end{cases}$$

↑
Key! Unlike a joint pdf / pmf, the likelihood fn. must also take in value(s) for the underlying parameter(s).

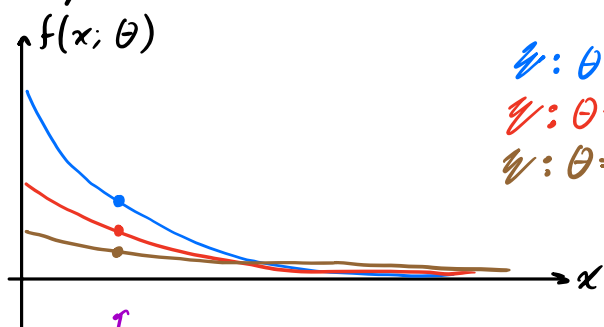
Intuition:

Each possible value of θ represents a different "universe" or "state of the world".

The likelihood fn. tells us how possible it is for each such "universe" (θ) to give rise to some data ($x_1, \dots, x_n$) we see.

Illustration: $n=1$, $X_1 \sim \text{Exp}(\theta)$, $\theta > 0$ unknown

Density of X : $f(x; \theta) = \theta \cdot e^{-\theta x}$



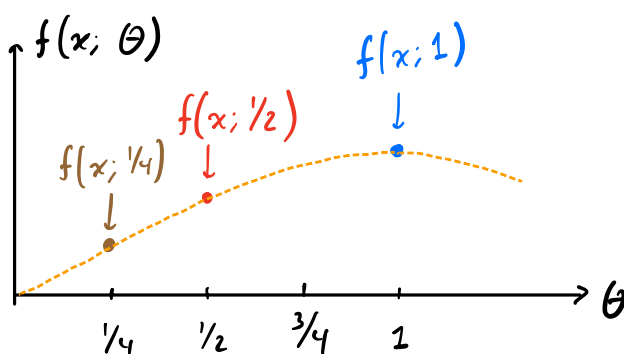
$\text{blue: } \theta = 1$

$\text{red: } \theta = 1/2$

$\text{brown: } \theta = 1/4$

(Example values of θ , all $\theta > 0$ are possible)

$\hookrightarrow$ Fix x , plot density (at fixed x) vs. θ



Once we fill this figure out for all possible θ , we get the likelihood fn.

$$L(\theta; x) = \theta \cdot e^{-\theta x}$$

Obs. #1: If $X_1, \dots, X_n$ are iid, with the marginal of X_i given by $f_x(x_i; \theta)$ (or $p_x(x_i; \theta)$) for unknown θ , then

$$L(\theta; x_1, \dots, x_n) = \underbrace{f_x(x_1; \theta)}_{p_x} \cdot \underbrace{f_x(x_2; \theta)}_{p_x} \cdot \dots \cdot \underbrace{f_x(x_n; \theta)}_{p_x}$$

(if rv's are discrete)

Obs. #2: If $X_1, \dots, X_n$ are discrete r.v.'s, then

$$L(\theta; x_1, \dots, x_n) = P(X_1 = x_1, \dots, X_n = x_n, \text{ given underlying param. equals } \theta)$$

Obs. #3: If we have $L(\theta^*; x_1, \dots, x_n) \geq L(\theta; x_1, \dots, x_n)$, then it would be more "likely" for us to see $X_1 = x_1, \dots, X_n = x_n$ with parameter θ^* , than with parameter θ .

$\hookrightarrow$ So, let's try to find what value of θ makes the data most likely!

Def. (Max. Likelihood Estimator/Estimate): Let $X_1, \dots, X_n$ be iid random variables with marginal $f(\cdot; \theta)$ (or $p(\cdot; \theta)$), for unknown value of θ . Then:

- The maximum likelihood estimator (MLE) for θ is:

$$\hat{\theta}_{MLE}(X_1, \dots, X_n) = \underset{\theta}{\operatorname{argmax}} \mathcal{L}(\theta; X_1, \dots, X_n)$$

$$\text{Random } X_1, \dots, X_n \Rightarrow \text{Random } \mathcal{L}(\theta; X_1, \dots, X_n) \Rightarrow \text{Random } \hat{\theta}_{MLE}$$

- Given realizations $x_1, \dots, x_n$, the max. likelihood estimate for θ is $\hat{\theta}_{MLE}(x_1, \dots, x_n)$.

Note: Unlike the previous section on unbiased estimators, we will not study in detail the distributional properties of $\hat{\theta}_{MLE}$. Still, we will mention some.

Properties of the MLE:

- $\hat{\theta}_{MLE}$ can be biased, but as $n \rightarrow \infty$, $\text{Bias}(\hat{\theta}_{MLE}, \theta) \rightarrow 0$.

I.e., $\hat{\theta}_{MLE}$ is asymptotically unbiased.

- As $n \rightarrow \infty$, the distribution of the MLE $\hat{\theta}_{MLE}(X_1, \dots, X_n)$ is approximately $N(\theta, \frac{1}{n} \text{?})$. I.e. $\hat{\theta}_{MLE}$ is asymptotically normal.

↑
Want to know what this is?
~ Take more stats!



Now, some examples!

Ex. 1: We have a coin with some unknown likelihood of giving "H" when flipped. After 10 flips, the first 8 came up "H", and the last 2 "T".

Q: What is the max. likelihood estimate for the prob. that the coin is "H"?

First, what family of distributions are we looking at?

→ Two distinct possible outcomes, so Bernoulli:

Next, how do we re-frame the setup through the lens of the Bernoulli distribution?

→ We had 10 iid random variables (the coin flips) coming from some Bernoulli distribution, where we do not know the success probability θ .

<u>Coinflip Outcome</u>		<u>Bernoulli Outcome</u>
H	$\longleftrightarrow$	1
T	$\longleftrightarrow$	0

So: $X_1, \dots, X_{10}$ $\overset{iid}{\sim}$ $\text{Ber}(\theta)$

↑ Not the observed H/T, but the random variables whose realizations we observed!

Now, what are the realizations we have of $X_1, \dots, X_{10}$?

→ First 8 flips were "H", so $x_1 = x_2 = \dots = x_8 = 1$

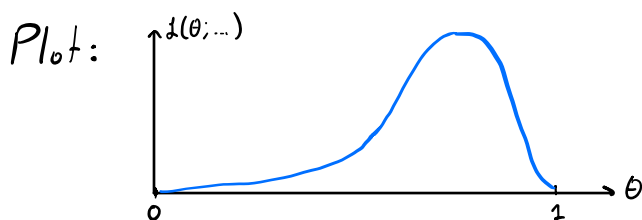
Last 2 were "T", so $x_9 = x_{10} = 0$

④ So, for any given θ , what is the likelihood function of our observed data, i.e. $L(\theta; 1, 1, \dots, 1, 0, 0) = ?$

$\begin{matrix} \uparrow & \uparrow & \uparrow & \uparrow & \uparrow \\ x_1 & x_2 & x_8 & x_9 & x_{10} \end{matrix}$

$\leadsto$ By independence, $L(\theta; \dots) = p(1; \theta) \cdot \dots \cdot p(1; \theta) \cdot p(0; \theta) \cdot p(0; \theta)$,
 where $p(x; \theta) = \begin{cases} \theta & \text{if } x=1 \\ 1-\theta & \text{if } x=0 \end{cases}$

$$\text{Hence, } L(\theta; \dots) = \theta \cdot \theta \cdot \dots \cdot \theta \cdot (1-\theta) \cdot (1-\theta) = \underline{\theta^8 (1-\theta)^2}$$



What value of θ maximizes the likelihood fn. (of our observed data)?

$$\begin{aligned} \leadsto \operatorname{argmax}_{\theta} L(\theta; \dots) &= \operatorname{argmax}_{\theta} \theta^8 (1-\theta)^2 \\ &= \operatorname{argmax}_{\theta} \log(\theta^8 (1-\theta)^2) \quad \left\{ \begin{array}{l} \text{since } \log(\cdot) \text{ is an} \\ \text{increasing function} \end{array} \right. \\ &= \operatorname{argmax}_{\theta} \underline{8 \log(\theta) + 2 \log(1-\theta)} \\ \frac{d}{d\theta} &= 8 \cdot \frac{1}{\theta} + 2 \cdot \frac{1}{1-\theta} \cdot (-1) \quad \left\{ \begin{array}{l} \text{set equal to 0} \\ \text{at } \theta = \hat{\theta}_{MLE} \end{array} \right. \\ 0 &= \frac{8}{\hat{\theta}_{MLE}} - \frac{2}{1 - \hat{\theta}_{MLE}} \\ \boxed{\hat{\theta}_{MLE} = \frac{8}{10}} & \quad \left\{ \begin{array}{l} \text{solving for } \hat{\theta}_{MLE} \end{array} \right. \end{aligned}$$

Ex. 1.1: Let's do the same problem, but without knowing the specific H/T we observed.

$\leadsto$ Amounts to not having $x_1 = x_2 = \dots = x_8 = 1, x_9 = x_{10} = 0$

What changes?

→ In step Ⓐ above, we can't plug in values for $x_1, \dots, x_{10}$
Instead, leave them as variables:

$$\mathcal{L}(\theta; \dots) = \underbrace{p(x_1; \theta)}_{\substack{\theta \text{ if } x_1 = 1 \\ 1 - \theta \text{ if } x_1 = 0}} \cdot \dots \cdot \underbrace{p(x_{10}; \theta)}_{\text{Same here!}}$$

$$= \theta^{\overbrace{\sum_{i=1}^{10} x_i}^{\text{the \# of } x_i\text{'s equal to 1}}} \cdot (1 - \theta)^{\overbrace{10 - \sum_{i=1}^{10} x_i}^{\text{the \# of } x_i\text{'s equal to 0}}}$$

Now, find the argmax over θ :

$$\rightarrow \underset{\theta}{\operatorname{argmax}} \theta^{\sum_{i=1}^{10} x_i} \cdot (1 - \theta)^{10 - \sum_{i=1}^{10} x_i} = \underset{\theta}{\operatorname{argmax}} (\sum_{i=1}^{10} x_i) \log(\theta) + (10 - \sum_{i=1}^{10} x_i) \log(1 - \theta)$$

Take $\log(\cdot)$ and simplify

Take a derivative w.r.t. θ , not w.r.t. $x_1, \dots, x_n$! Set this equal to 0 at $\theta = \hat{\theta}_{MLE}$:

$$0 = \frac{\sum_{i=1}^{10} x_i}{\hat{\theta}_{MLE}} - \frac{10 - \sum_{i=1}^{10} x_i}{1 - \hat{\theta}_{MLE}}$$

$$\Rightarrow \boxed{\hat{\theta}_{MLE}(x_1, \dots, x_{10}) = \frac{1}{10} \sum_{i=1}^{10} x_i}$$

Note: In the case where $X_1, \dots, X_n \stackrel{iid}{\sim} \text{Ber}(\theta)$, for unknown θ , it will be the case that $\hat{\theta}_{MLE}(X_1, \dots, X_n) = \frac{1}{n} \sum_{i=1}^n X_i$.

The same as Ex. 1.1, but before we actually observe realizations $x_1, \dots, x_n$.

Ex. 2: Let $X_1, \dots, X_n \stackrel{iid}{\sim} N(\mu, 1)$, with unknown μ .

a) If $n=1$, and we observe $X_1 = x_1$, then what is $\hat{\theta}_{MLE}(x_1)$?

$$\hat{\theta}_{MLE}(x_1) = \underset{\theta}{\operatorname{argmax}} f_x(x_1; \theta, 1) \parallel \text{The pdf of a } N(\theta, 1) \text{ r.v.}$$

$$= \underset{\theta}{\operatorname{argmax}} \frac{1}{\sqrt{2\pi}} e^{-\frac{(x_1 - \theta)^2}{2}} \quad \left. \vphantom{\frac{1}{\sqrt{2\pi}}} \right\} \text{Drop the constants}$$

$$= \underset{\theta}{\operatorname{argmax}} e^{-\frac{(x_1 - \theta)^2}{2}} \quad \left. \vphantom{e^{-\frac{(x_1 - \theta)^2}{2}}} \right\} \text{Take log}$$

$$= \underset{\theta}{\operatorname{argmax}} -\frac{(x_1 - \theta)^2}{2} \quad \left. \vphantom{-\frac{(x_1 - \theta)^2}{2}}} \right\} \text{Drop constants}$$

$$= \underset{\theta}{\operatorname{argmax}} \underline{-(x_1 - \theta)^2}$$

$$\frac{d}{d\theta} = \underline{-2(x_1 - \theta)(-1)}$$

equals 0 at $\theta = x_1$

$$\boxed{\hat{\theta}_{MLE}(x_1) = x_1}$$

b) If $n > 1$ and we observe realizations $x_1, \dots, x_n$, what is $\hat{\theta}_{MLE}(x_1, \dots, x_n)$?

$$\hat{\theta}_{MLE}(x_1, \dots, x_n) = \underset{\theta}{\operatorname{argmax}} \prod_{i=1}^n \frac{1}{\sqrt{2\pi}} e^{-\frac{(x_i - \theta)^2}{2}}$$

$$= \underset{\theta}{\operatorname{argmax}} (2\pi)^{-n/2} \cdot e^{-\frac{1}{2} \sum_{i=1}^n (x_i - \theta)^2}$$

$$= \underset{\theta}{\operatorname{argmax}} \underline{-\sum_{i=1}^n (x_i - \theta)^2}$$

$$\frac{d}{d\theta} = \underline{-2 \sum_{i=1}^n (x_i - \theta)(-1)}$$

equals 0 at $\theta = \frac{1}{n} \sum_{i=1}^n x_i$

$$\boxed{\hat{\theta}_{MLE}(x_1, \dots, x_n) = \frac{1}{n} \sum_{i=1}^n x_i}$$

← MLE for unknown mean of a normally distributed pop. is the sample mean!